

Reviewer-response action matrix

29 September 2026. Completed software work is distinguished from experiments and author information still needed. This is a remediation tracker, not a claim that all rejection reasons are resolved.

Reviewer 1 item	Action / evidence	Status before submission
1. Empty images	Explicit both-empty Dice=1, one-empty Dice=0; masks decoded by nonzero labels; numerical tests; foreground and empty strata in the new validation study.	Software complete. BBBC039 validation contains only one empty reference image, so add independent true empty fields.
2. Source-volume independence	Public official split manifest plus image/group/hash disjointness.	Public split checked. Private source-volume identifiers still unresolved; do not claim private leakage-proof validation.
3. Reproducibility	New code repository, locked configs, download script, result CSVs, checkpoint hashes.	Local reproduction supported; repository initially private and weights local. Public code/weights release and inherited licensing still needed.
4. Tversky interpretation	New study uses explicit Dice+BCE, no Tversky or unsubstantiated recall-bias statement.	Resolved for new study. Historical FP/FN coefficient interpretation documented in audit.
5. Boundary claims	Explicit differentiable morphological boundary Dice term; native-grid BF1, surface Dice, HD95.	Implemented and tested; empirical usefulness must follow measured ablation results.
6. Gabor robustness	Earlier corrected private-data runs include orientation/scale/sigma sweeps and edge controls. A frozen 30-run public validation-only study repeats orientation, scale, sigma, Sobel and neutral-edge controls over three seeds.	Completed 29 September 2026. Effects are under 0.15 Dice points and parameter-sensitive. It remains exploratory because BBBC039 informed development; independent confirmation still remains.
7. Unequal budgets	Same 100 training fields, 50 validation fields, 20 epochs, optimizer, schedule, augmentation and threshold for new comparisons.	Completed for new study; architecture-dependent compute differs and should be reported. Historical comparisons remain unequal.
8. Sampling details	Formula in core.py, binary-encoding and enable/disable regression tests. Uniform sampling throughout the first public studies; the frozen validation study includes a matched weighted-versus-uniform condition.	Completed: +0.0473 foreground Dice points, but the sole empty case was correctly empty in only one of three weighted runs. Independent confirmation remains.
9. Zero-positive batches	Safe optional adaptive weighting; tests on all-zero, all-one and extreme-logit batches.	Software complete. Main public objective uses unweighted BCE.
10. Annotation quality	Public annotations attributed to providers.	Authors must establish original private annotators, protocol, inter-rater agreement and uncertainty; cannot fabricate retrospectively.
11. Clinical claims	No clinical-grade, diagnostic or deployment claim in the new draft.	Resolved by narrowing scope; no clinical validation performed.
12. Statistics	Per-image scores, descriptive bootstrap CIs, paired comparisons with Holm correction; patient aggregation for TNBC.	Exploratory inference; BBBC039 has one acquisition study, not 50 independent patients. New independent confirmation still needed.
13. Quantum/AI figures	Actual image/prediction plots; no quantum mechanism claimed. AI assistance disclosure drafted.	Author review of disclosure and all figure attributions required.
14. External validation	Complete frozen-checkpoint BBBC038, BBBC039 and TNBC partitions; all new models transferred to TNBC without tuning.	Measured evidence includes failures. BBBC038/039 overlap and foundation-model training overlap prevent broad claims.

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15. TTA	Fixed threshold 0.5, no TTA for any new neural-model comparison.	Complete; separate TTA experiments would need matched settings.
16. Small data/overfit	Three seeds, full learning curves, ImageNet initialization, AdamW and geometrical augmentation.	Partial: official public holdout is not patient-level cross-validation; add independent data-efficiency curves and grouped CV.
17. B7 contribution	Vanilla B7, SCSE-only B7 and controlled edge/loss variants. Earlier private encoder ladder (B0/B3/B5/ResNet34/50) is retained.	Public replication of smaller-encoder controls and modern matched-data baselines remains needed. Do not discard the already completed private study.
18. Crops/ROI	No ROI or crop generation in new public study; complete fields resized to 512 and probabilities restored to native grid.	New protocol complete; private historical crop/source provenance still needed if reported.
19. Inconsistent IoU	Historical arithmetic audit: stored JSON means exceed saved values by 0.01 for Dice and IoU. New tables generated from CSVs.	Do not reuse historical inflated means. Generated new numbers independently rechecked.
20. Failed baselines	All 24 supplied historical checkpoints attempted, including both export versions; failures retained explicitly. New run outputs not discarded for poor results.	Check evaluation failure log; historical failures are not evidence of novel-method superiority.

Reviewer 2 and EMBC concerns

The draft focuses its introduction on the empirical question, adds directly relevant segmentation literature, defines the nucleus-foreground target, provides exact dataset/task descriptions, removes quantum framing and synthetic science illustrations, and distinguishes method components through controlled comparisons. Numerical tables come from measured outputs. Unrelated EEG, depression or temporal-text papers suggested in a review are not added as token citations: relevance must be justified scientifically. Histopathology classification papers may motivate the wider application but are not segmentation baselines.

The most important outstanding issue is a defensible contribution beyond combining established blocks. If the edge branch does not consistently improve matched baselines, a superiority paper is not justified. A transparent evaluation paper may still be useful, but a strong methods submission needs either a validated improvement, a distinctive data-efficiency/robustness result, or a reusable dataset/protocol contribution.